Design, Development, and Evaluation of a Smart Facility Measurement System for Indoor Lighting and Air-Conditioning and Their Energy Efficiency

A Research Proposal

Presented to

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Chapter 1

Introduction

Background and Rationale

Buildings account for a significant portion of global energy consumption, making energy efficiency one of the primary concerns in modern facility management. According to the International Energy Agency (IEA), the adoption of digital technologies, smart measurement systems, and data-driven energy management strategies has become increasingly important in reducing energy consumption, improving operational efficiency, and promoting sustainable building operations. Continuous measurement and real-time analysis of building systems enable facility managers to identify inefficient equipment operation, optimize energy usage, and support timely maintenance decisions. As educational institutions continue to expand their facilities and increase their dependence on electrical and mechanical systems, the implementation of intelligent measurement technologies has become essential in achieving sustainable and energy-efficient building management.

Recognizing the importance of energy conservation, governments have established policies that encourage institutions to adopt energy-efficient technologies and improve facility management practices. In the Philippines, this commitment is reflected through the implementation of Republic Act No. 11285, otherwise known as the Energy Efficiency and Conservation Act, and its corresponding Implementing Rules and Regulations issued under Department Circular No. DC2019-11-0014 by the Department of Energy (DOE). The policy encourages institutions to establish energy management practices, adopt energy-efficient technologies, and continuously measure building energy consumption to improve operational efficiency and reduce unnecessary energy use. These national initiatives emphasize that

effective energy management requires continuous measurement of the major building systems responsible for energy consumption

Among the various electrical and mechanical systems operating within buildings, lighting systems and heating, ventilation, and air-conditioning (HVAC) systems account for a significant portion of overall energy consumption. Proper indoor lighting is essential for providing visual comfort, improving productivity, and ensuring a safe learning environment, while HVAC systems maintain acceptable indoor thermal conditions that directly influence occupant comfort and building energy use. Because of their substantial contribution to building energy demand, both systems require regular assessment to ensure efficient operation, minimize unnecessary energy consumption, and maintain a comfortable indoor environment. Consequently, several organizations have developed standardized procedures and performance criteria for evaluating these systems.

For indoor lighting, the National Renewable Energy Laboratory (NREL) recommends systematic multi-point illumination measurements to accurately evaluate lighting conditions, while EN 12464-1 establishes the recommended illumination levels for educational and commercial facilities. Failure to maintain appropriate lighting levels may result in inadequate visual performance, occupant discomfort, and unnecessary electrical energy consumption. Similarly, ANSI/ASHRAE Standard 55 provides the recommended thermal comfort criteria for occupied indoor spaces to ensure that HVAC systems maintain acceptable indoor environmental conditions. These standards serve as important references for assessing facility performance; however, conventional evaluation methods in many institutions remain dependent on manual measurements and periodic inspections, making it difficult to continuously measure facility conditions and immediately identify inefficient system operation.

Recent advances in Internet of Things (IoT) technology have addressed these limitations by enabling the development of low-cost and portable measuring systems capable of continuously collecting environmental and electrical data. Studies conducted by Zafar et al.

(2018) and Yang et al. (2025) demonstrated that Arduino-based sensor networks integrated with wireless communication technologies provide an effective platform for real-time environmental measurement and centralized data visualization. Likewise, Parkinson et al. (2019) emphasized that continuous measurment using distributed sensor networks provides a more representative evaluation of indoor environmental conditions than conventional single-point or periodic measurements, thereby improving the reliability of facility assessment. These developments demonstrate that continuous sensor-based measuring offers a practical approach for improving building management and supporting data-driven decision-making.

Beyond continuously measuring environmental conditions, recent studies have highlighted the importance of evaluating the operational performance and energy efficiency of HVAC systems using measured operating parameters. Rønneseth et al. (2018) emphasized that accurate airflow measurement is fundamental in assessing HVAC performance and estimating energy efficiency, while Cline (2010) demonstrated that cooling capacity and energy efficiency can be estimated through air-side measurements involving airflow, temperature, and electrical parameters. Furthermore, Taal (2021) highlighted that continuous energy performance assessment enables the early identification of inefficient HVAC operation and supports preventive maintenance through automated performance evaluation. These studies demonstrate that combining continuous environmental measuring with performance analysis provides a more comprehensive approach to facility assessment than conventional condition measuring alone.

Despite these technological developments, most existing measuring systems focus only on individual aspects of facility management, such as indoor lighting, environmental measuring, or HVAC systems independently. Likewise, many commercially available solutions require expensive building automation infrastructure or specialized equipment that may not be practical for educational institutions and small- to medium-scale facilities. Consequently, there remains a need for a portable, wireless, and integrated measuring system capable of simultaneously

assessing indoor lighting conditions and evaluating the operational performance and energy efficiency of air-conditioning systems while automatically generating recommendations for facility improvement and preventive maintenance.

In response to this need, this study proposes the design, development, and evaluation of a Smart Facility Measuring System capable of integrating indoor lighting assessment and air-conditioning energy efficiency analysis into a single microcontroller-based platform. The proposed system utilizes wireless sensor technology to acquire illumination, electrical, thermal, and airflow measurements, which are processed through a centralized mobile and desktop application to perform real-time measuring, estimate HVAC energy efficiency, assess lighting conditions based on established standards, and generate automated recommendations. By providing an accessible and data-driven facility assessment tool, the study aims to support energy-efficient building management, improve maintenance planning, and contribute to the sustainable operation of educational institutions and other indoor facilities.

Objectives of the Study

General Objective

To design, develop, and evaluate a Smart Facility Measuring System for indoor lighting measuring and air-conditioning energy efficiency evaluation.

Specific Objective

Specifically, this study seeks to:

1. Design a Smart Facility Measurement and Assessment System that integrates sensor technology, microcontroller-based hardware, and a mobile and desktop application for measuring indoor lighting and air-conditioning performance.

2. Develop an Indoor Lighting Measurement Module capable of acquiring illumination measurements from multiple locations within an indoor facility and assessing compliance with established lighting standards.
3. Develop an Air-Conditioning Measurement Module capable of acquiring electrical and thermal operating parameters for evaluating the energy efficiency and indoor thermal environment of an air-conditioning system.
4. Evaluate the Smart Facility Measurement and Assessment System through a Flutter-based mobile and desktop application by assessing its functionality, accuracy, reliability, and to generate automated assessment results and recommendations.

Significance of the Study

Facility Management Personnel. The developed Smart Facility Measurement and Assessment System will provide facility management personnel with a reliable and convenient tool for measuring indoor lighting and air-conditioning conditions within building facilities. The system will assist in identifying lighting deficiencies and air-conditioning performance issues through real-time measurement and assessment, enabling timely maintenance decisions, preventive maintenance planning, and facility improvement initiatives. Furthermore, it will reduce the time and effort required for manual inspections while supporting efficient facility operation and maintenance.

Educational Institutions. Schools, colleges, and universities may utilize the system to assess classrooms, laboratories, faculty offices, libraries, and other learning spaces to help maintain a safe, comfortable, and conducive learning environment.

Commercial and Government Facilities. Office buildings, commercial establishments, hospitals, and government institutions may use the developed system to support routine facility inspections and improve operational efficiency.

Future Researchers. This study may serve as a reference for future studies involving smart facility measuring, wireless sensor networks, embedded systems, Internet of Things (IoT), and indoor environmental assessment.

Scope and Limitations

This study focuses on the design, development, and evaluation of a Smart Facility Measuring System for indoor lighting and air-conditioning in various indoor facilities. The system consists of two measuring modules: a lighting module that measures illumination levels at multiple locations within a room using wireless sensors, and an air-conditioning module that measures the electrical and thermal parameters of an air-conditioning unit to assess its operational performance.

Both modules are integrated into a mobile and desktop application that performs real-time data acquisition, automated computation, assessment, and recommendation generation based on applicable standards. The developed prototype is intended for use in various types of indoor facilities such as classrooms, laboratories, faculty offices, administrative offices, and other similar environments where lighting and air-conditioning systems are present.

However, the study is limited to the measuring and energy-efficiency of indoor lighting and air-conditioning systems and does not include automatic control or operation of these systems. The prototype evaluates only the parameters measured by the installed sensors and user-provided facility information and does not assess other indoor environmental factors such as humidity, indoor air quality, noise levels, or occupancy. Furthermore, the developed system is intended as a decision-support tool for facility measuring, energy-efficiency assessment, and maintenance planning and is not designed to replace comprehensive professional energy audits or engineering inspections.

Chapter 2

Review of Related Literature

This chapter presents the review of related literature and studies that provide the theoretical and technical foundation for the development of the proposed Smart Facility Measuring System. The reviewed literature covers concepts related to energy efficiency, indoor lighting measuring, air-conditioning performance evaluation, wireless sensor technology, Internet of Things (IoT)-based measuring systems, and automated facility assessment. These references establish the current practices, technologies, and methodologies used in measuring energy-related parameters and demonstrate the importance of real-time data acquisition and analysis in improving facility energy management.

Furthermore, the reviewed studies identify existing limitations and research gaps that support the need for the development of a portable measuring system capable of integrating indoor lighting and air-conditioning energy performance evaluation into a single platform. The findings from these related works serve as the basis for the design, development, and evaluation of the proposed system.

The reviewed literature is organized according to the main technical areas of the study. It starts with the principles of energy efficiency, digitalization, and continuous measuring. Literature on indoor lighting lays the groundwork for measuring lighting and evaluating energy performance. Studies on air-conditioning systems outline the principles for assessing thermal, airflow, electrical, and cooling performance. In addition, related works on wireless sensor networks, IoT-based measuring, and centralized data collection provide the basis for merging the lighting and air-conditioning systems into one smart facility measuring system.

Internet-Based Measuring and Controls for HVAC Applications

Wang and Burnett (2001) proposed an Internet-based measuring and control architecture for Heating, Ventilation, and Air-Conditioning (HVAC) systems. The architecture integrates sensors, data acquisition modules, communication interfaces, web-based applications, and databases into a centralized measuring framework. The system enables remote acquisition of HVAC operating parameters, real-time measuring, and improved facility management through Internet connectivity. The study emphasized the importance of integrating measuring, communication, and data processing into a unified platform to improve operational efficiency and support timely maintenance activities.

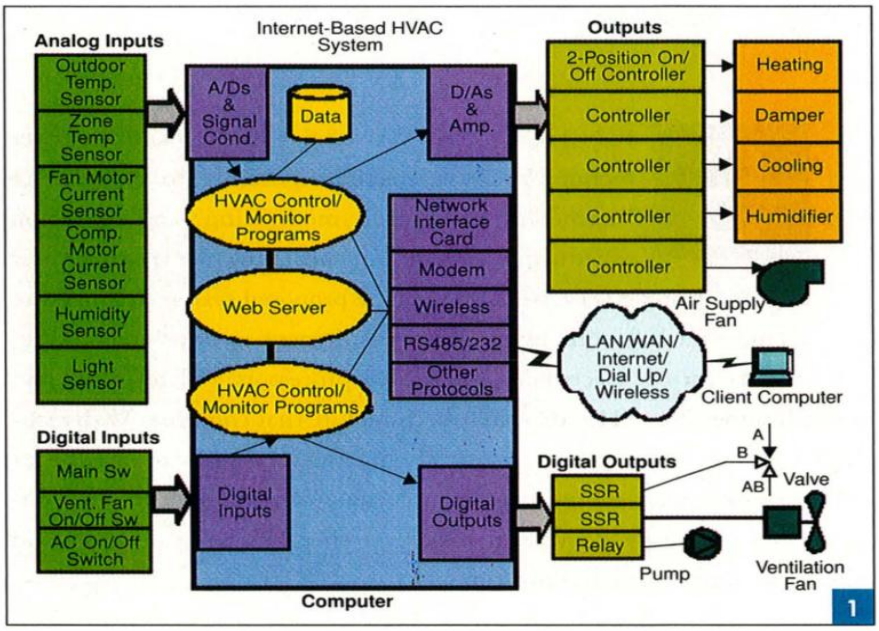


Figure 1

Internet-Based HVAC Measuring and Control Architecture

Figure 1 shows the Internet-based HVAC measuring and control architecture proposed by Wang and Burnett (2001). The architecture integrates sensors and actuators, data acquisition and signal conditioning, input/output interfaces, networking components, control

programs, a web server, and a database within a client/server framework. HVAC system data are acquired through the sensing and data acquisition components and communicated through the network for processing, storage, and remote access. The architecture provides an integrated framework for real-time measuring, data management, and web-based interaction with HVAC systems.

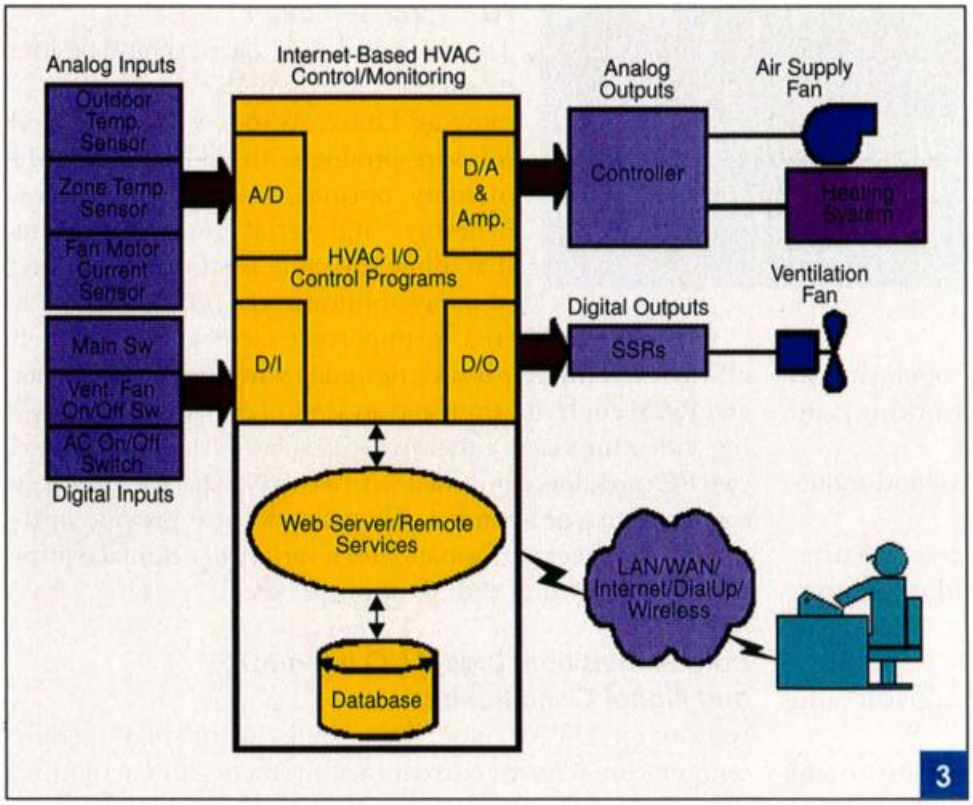


Figure 2

Experimental Internet-Based HVAC Measuring System

Figure 2 presents the experimental Internet-based HVAC measuring and control system developed by Wang and Burnett (2001). The system integrates computer-based data processing, web server and database functions, digital input/output hardware, electrical

measurement equipment, and Ethernet-based input/output modules. These components enable the acquisition of HVAC system measurements and the exchange of measuring and control information through a networked platform.

Arduino-Based IoT Smart Building Measuring System

HAR et al. (2018) developed an Arduino-based Internet of Things (IoT) measuring system designed for smart building applications. The proposed system utilized multiple environmental sensors connected to an Arduino microcontroller to acquire real-time environmental data. The collected measurements were transmitted wirelessly to a measuring platform, allowing users to remotely measure building conditions through an Internet-connected application. The study demonstrated that integrating wireless communication with microcontroller-based sensor networks improves accessibility, measuring efficiency, and data availability in smart building environments.

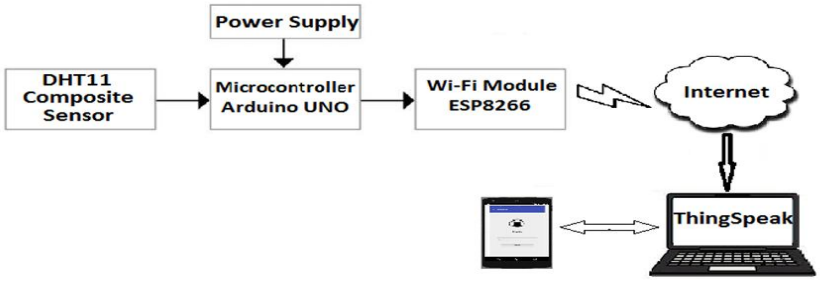


Figure 3

Hardware Architecture of an IoT-Based Measuring System

Figure 3 presents the hardware architecture of the IoT-based environmental measuring system developed by Zafar et al. (2018). The DHT11 sensor acquires temperature and humidity measurements and transmits the collected data to an Arduino UNO microcontroller. The

microcontroller processes the sensor readings and transfers the data through an ESP8266 Wi-Fi module over the Internet to the ThingSpeak cloud platform for data storage, analysis, and visualization.

ESP32-Based Wireless Sensor Networks for Smart Facility Measuring

Wireless Sensor Networks (WSNs) have become an essential technology in smart building and facility measuring because they enable the collection, transmission, and management of environmental data from multiple sensing locations through wireless communication. Unlike conventional measuring systems that rely on extensive wired connections, WSNs utilize distributed sensor nodes capable of communicating with a centralized measuring platform, thereby improving installation flexibility, reducing wiring complexity, and facilitating system scalability. The integration of WSNs in intelligent buildings has supported various applications, including indoor environmental measuring, energy management, occupancy detection, and facility automation, making them an effective solution for real-time measuring and assessment of building conditions.

Khan et al. (2022) developed an ESP32-based wireless sensor network utilizing the ESP-MESH communication protocol for real-time environmental measuring. The proposed system consisted of multiple ESP32 microcontroller-based sensor nodes integrated with environmental sensors to collect data from different measuring locations. Each sensor node was capable of wirelessly transmitting the collected measurements to a centralized measuring platform through the ESP-MESH network architecture, enabling continuous measuring and centralized data management. The use of the ESP32 microcontroller provided integrated Wi-Fi communication, sufficient processing capability, and compatibility with various environmental sensors, making it suitable for implementing scalable wireless measuring systems in smart building applications.

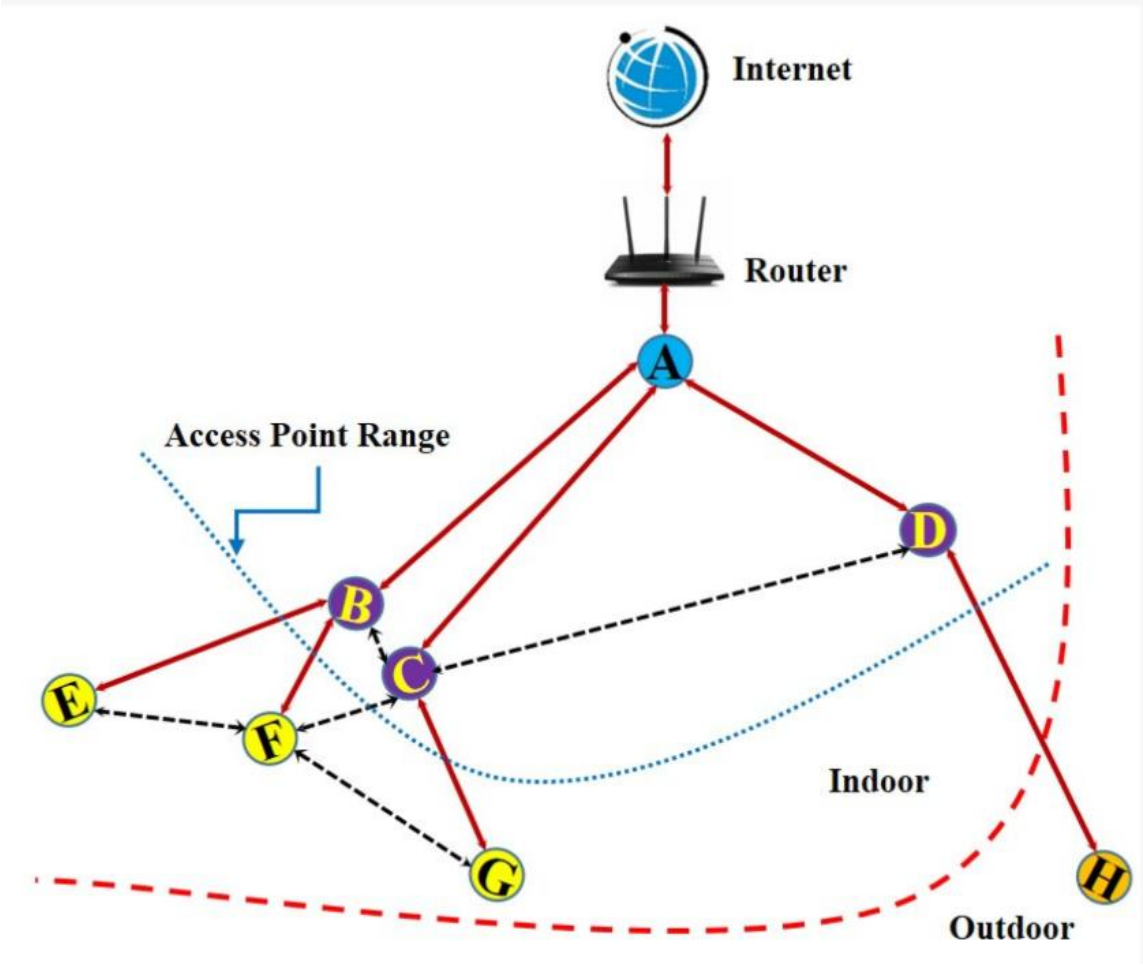


Figure 4

ESP-MESH Wireless Sensor Network Topology

Figure 4 illustrates the ESP-MESH wireless sensor network topology presented by Khan et al. (2022). Figure 1(a) shows the limitations of a conventional Wi-Fi network, where communication depends on a direct connection between the router and sensor nodes within the access point range. In contrast, Figure 1(b) demonstrates the ESP-MESH architecture, where intermediate nodes relay data between neighboring sensor nodes, extending network coverage beyond the access point's range. This self-organizing communication enables multiple ESP32-based sensor nodes to transmit data reliably to a centralized measuring platform, making the architecture suitable for scalable smart facility measuring applications.

Arduino-Based Indoor Environmental Measuring System

Yang et al. (2025) developed an Arduino-based sensor network for indoor environments that integrates multiple sensors to improve energy efficiency and environmental safety. The system utilizes a light intensity sensor to automatically regulate artificial lighting based on available natural daylight, while temperature and humidity sensors measure indoor environmental conditions to control fan operation. A flame sensor was also incorporated to enhance indoor safety by activating alarms when fire hazards are detected. Through the integration of multiple sensor modules and automated control, the system demonstrated the capability of maintaining indoor comfort while reducing unnecessary energy consumption using a low-cost and scalable measuring platform.

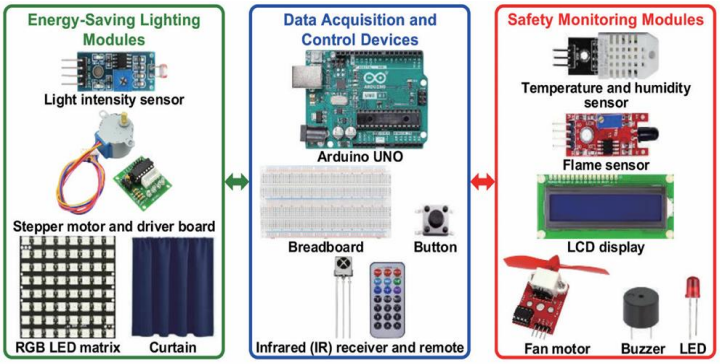


Figure 5

Arduino-Based Multi-Sensor Measuring System

Figure 5 illustrates the Arduino-based multi-sensor indoor environmental measuring system developed by Yang et al. (2025). The architecture integrates multiple environmental sensors, including light intensity, temperature, humidity, and flame sensors, through a centralized microcontroller to perform real-time data acquisition and automated measuring. The modular design demonstrates how different sensing devices can operate as a unified measuring platform capable of supporting lighting control, environmental measuring, and safety-related

functions. This integration highlights the feasibility of implementing a centralized sensor network for indoor facility measuring, providing a conceptual basis for developing multi-parameter measuring systems.

ESP32 and Flutter-Based Mobile Measurement Systems

Ferreira et al. (2026) developed an Internet of Things (IoT)-based mobile measurement and control system that integrated an ESP32 microcontroller, a Flutter-based mobile application, and wireless communication technologies for remote interaction with a laboratory water level system. The study evaluated the capability of the ESP32 to acquire sensor measurements while communicating with a Flutter application through MQTT and Bluetooth Low Energy (BLE), enabling real-time data visualization and remote operation. The authors highlighted that the ESP32's built-in wireless capabilities and dual-core architecture, combined with Flutter's cross-platform development framework, provided a reliable and responsive platform for engineering applications requiring sensor data acquisition and user interaction. Furthermore, the proposed system demonstrated stable communication with low latency, efficient wireless data transmission, and an intuitive graphical user interface, illustrating the effectiveness of integrating ESP32 and Flutter for IoT-based engineering systems.

BH1750 and DHT22 Sensors for Indoor Environmental Measurement

Desnanjaya et al. (2022) developed an ESP-12E-based room monitoring system that utilized the BH1750 digital light sensor to measure indoor illumination and the DHT22 sensor to measure room temperature and humidity. The study demonstrated that the BH1750 provides direct digital illuminance measurements in lux through an I²C communication interface without requiring complex signal processing, while the DHT22 accurately acquires indoor temperature and humidity data for environmental assessment. The authors further emphasized that the BH1750 has a wide measurement range of 0 to 65,535 lx and is capable of measuring illumination from various light sources, including sunlight, incandescent lamps, fluorescent

lamps, and LEDs. The successful integration of the BH1750 and DHT22 with an ESP-based microcontroller demonstrated their suitability for low-cost indoor environmental measurement systems and building condition assessment applications.

Lighting Measurement Procedures for Commercial Buildings

Deru et al., (2005), through the National Renewable Energy Laboratory (NREL) developed standardized procedures for measuring indoor lighting performance in commercial buildings to support lighting quality assessment and energy management. The guideline recommends obtaining illumination measurements from multiple locations within a room using calibrated light-measuring instruments to accurately represent the overall lighting condition of an indoor facility. It also emphasizes comparing the measured illumination levels with established lighting standards to identify areas that are underlit, adequately illuminated, or overlit. These procedures provide a systematic approach for evaluating lighting performance and identifying opportunities for lighting improvement.

Performance Metric	Data Needed	Measurement Location or Source of Data
Building Energy Use	total building energy use	
Code Baseline Lighting Energy	**Code Baseline LPD**, *lighting schedules*, and *lighted floor area*	
Code Baseline LPD	function of each space and LPDs for each space from the applicable energy code	
Daylighting and Occupancy Energy Savings	**Lighting Energy without Controls** and **Lighting Energy with Daylighting**	
Global Horizontal External Illuminance	measured external illuminance	
Horizontal Illuminance	measured interior horizontal illuminance at several points	
Installed Baseline Lighting Energy	**Lighting Energy Use** with occupancy sensors disabled and daylighting controls turned off – measured or estimated with **Installed Lighting Power** and *lighted hours*	
Installed Lighting Power	**Installed LPD** and *lighted floor area*	
Installed LPD	Installed lighting capacity (W) in each space by type (ambient, accent, etc.) and lighted floor area for each space	
Instantaneous Heat Gain from Lighting	Number and types of fixtures, lighting power by fixture type, and fixture power ratios for each lighting type	
Lighting Design Energy Savings	**Code Baseline LPD** and **Installed LPD**	
Lighting Energy Cost	Time-series energy data for **Lighting Energy Use** and **Building Energy Use**, the lighting contribution to electrical demand charge, and the utility rate structure	
Lighting Energy Percent of Building Energy	**Lighting Energy Use**, **Building Energy Use**	
Lighting Energy Use	Tier 1: Fixture type, power per fixture, fixture power ratio, fixture counts for all lighting types, and *lighting schedules* Tier 2: Measured monthly **Lighting Energy Use**	
Peak Lighting Power	Subhourly lighting energy data	
Plug-in Lighting Energy Use	Estimated energy use of plug-in lighting (see Section 6.5.2)	
Total Lighting Energy Savings	**Code Baseline LPD**, measured lighting schedule, *lighted floor area*, and **Lighting Energy Use**	
Vertical Illuminance	measured vertical illuminance	

Figure 6

Lighting Performance Metrics and Required Data

Figure 6 presents the performance metrics and corresponding data required for evaluating indoor lighting energy performance. The procedure identifies horizontal illuminance as a lighting performance metric measured at several interior points. It also includes lighting energy use, installed lighting power, lighting power density, and energy savings as relevant parameters for assessing the energy performance of indoor lighting systems.

Smart Lighting Systems for Classroom Environments

Sun et al. (2020) developed a self-optimizable smart lighting system designed to improve classroom illumination by automatically adjusting brightness, correlated color temperature (CCT), and illuminance distribution based on different learning contexts. The system integrates Internet of Things (IoT) technology, ambient light sensors, occupancy sensors, and cloud-based data processing to continuously measure classroom conditions and automatically switch between lighting modes according to scheduled classroom activities. In addition to enhancing students' visual comfort and learning performance, the system also improves energy efficiency through intelligent lighting control and continuous environmental measuring. The study demonstrated that integrating sensor technologies with automated lighting control provides a flexible and adaptive approach to managing indoor lighting environments.

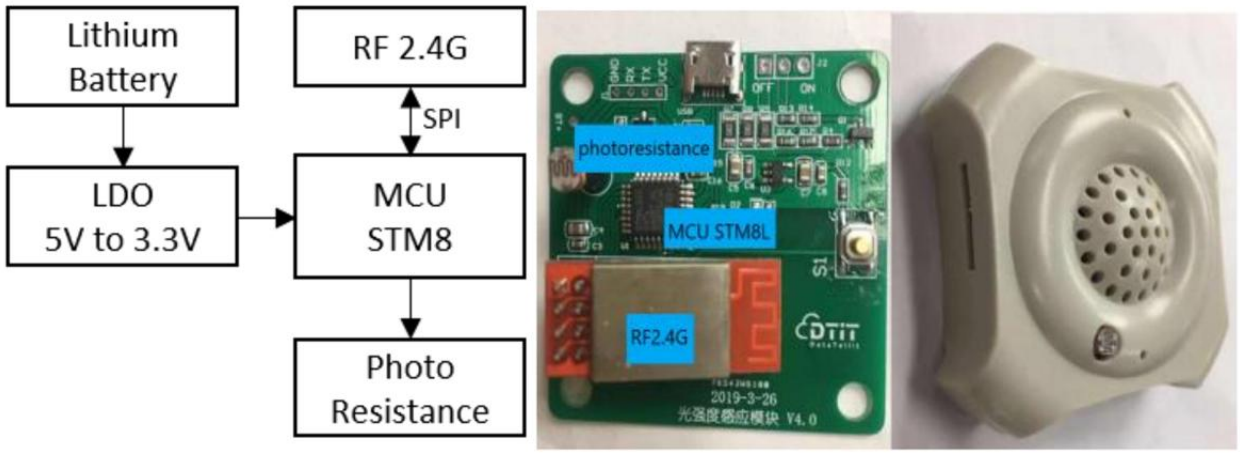


Figure 7

Ambient Light Sensor Module

Figure 7 presents the ambient light sensor module developed by Sun et al. (2020) for measuring luminous levels within specific classroom zones. The module integrates a microcontroller, a wireless RF communication component, a battery power supply, and a

photoresistor for ambient light sensing. The photoresistor generates sensor data that are sampled through the microcontroller's analog-to-digital conversion input and transmitted through the wireless communication module. This architecture enables the acquisition and wireless transmission of lighting data for continuous measuring.

Lux Sensor Coverage and Multi-Point Illumination Measurement

Deru et al. (2005) developed a standardized procedure for measuring indoor lighting performance in commercial buildings through the National Renewable Energy Laboratory (NREL). The study explains that indoor illumination is not uniformly distributed across a room due to variations in luminaire placement, room geometry, and surface reflectance. To obtain representative lighting data, the researchers recommended dividing the room into a measurement grid and collecting illuminance readings from multiple locations across the occupied area before calculating the average illuminance. The procedure emphasizes that a single measurement point cannot accurately represent the overall lighting condition of an indoor environment and that multi-point measurement provides a more reliable basis for evaluating indoor lighting performance.

Airflow Measurements for Air Handling Units

Rønneseth et al. (2018) conducted a study on airflow measurement techniques for air handling units (AHUs) to evaluate the field performance and energy efficiency of Heating, Ventilation, and Air-Conditioning (HVAC) systems. The researchers performed a technical investigation of various airflow measurement methods, including pressure differential, velocity traversal, ultrasonic, and tracer gas techniques, while reviewing applicable standards and conducting laboratory and field evaluations of airflow measuring equipment. The study found that airflow measurement is a fundamental requirement for assessing HVAC energy efficiency because airflow directly influences cooling performance, thermal comfort, and system operation.

Among the evaluated techniques, the velocity traversal method using an anemometer was identified as a practical approach for field measurements, although multiple measurement points are necessary to improve accuracy under actual operating conditions. The study further emphasized that accurate airflow measurements enable the identification of installation errors, operational faults, and performance degradation, thereby supporting more effective energy management and preventive maintenance of HVAC systems.

Psychrometric Testing Facility Restoration and Cooling Capacity Testing of a Residential Split Air Conditioner

Cline (2010) conducted a study on the restoration of a psychrometric testing facility to evaluate the cooling capacity and energy efficiency of a residential split-type air-conditioning system in accordance with industry testing standards. The researcher utilized an experimental methodology by instrumenting the air-conditioning system with temperature sensors, airflow measuring devices, electrical measuring instruments, and a computerized data acquisition system to obtain the thermal and electrical parameters required for performance evaluation. The study employed the air-side enthalpy method to determine the cooling capacity of the air-conditioning unit and compared the measured performance with manufacturer ratings and applicable AHRI and ASHRAE standards. The results demonstrated that accurate measurements of airflow, air temperature, and electrical power are essential for determining cooling capacity and evaluating the energy efficiency of air-conditioning systems. Furthermore, the study emphasized that standardized testing procedures and continuous data acquisition improve the reliability of HVAC performance evaluation and facilitate the identification of performance deviations.

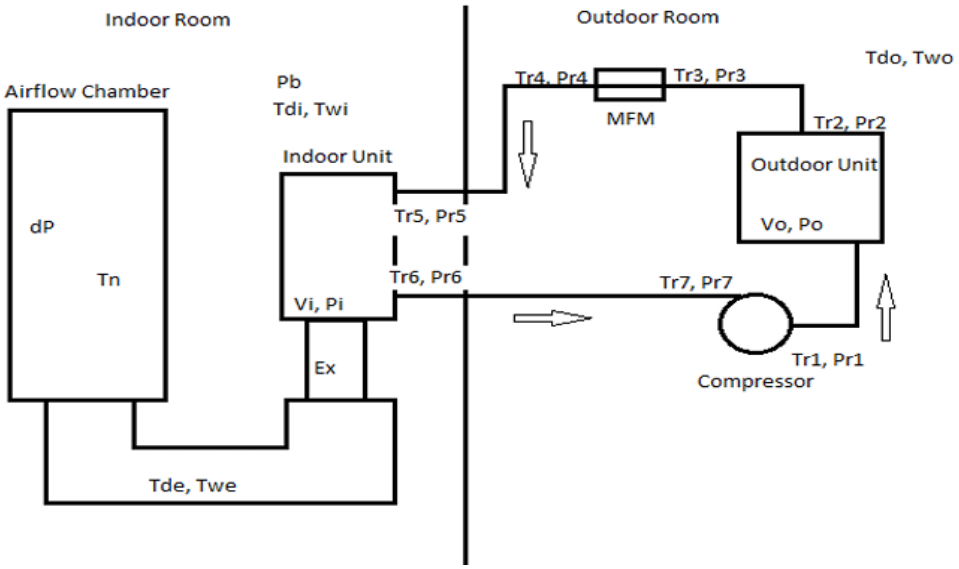


Figure 8

Air-Conditioning System Data Collection Locations

Figure 8 presents the refrigeration system layout and the data collection locations used by Cline (2010) during cooling-capacity testing. The measurement points are distributed across the indoor and outdoor sections of the air-conditioning system to acquire thermal, electrical, airflow, pressure, and refrigerant parameters. The collected measurements are integrated into a data acquisition system and used in calculating cooling capacity and evaluating system performance.

Automated Energy Performance and Fault Detection of HVAC Systems

Taal (2021) developed the 4S3F Method, an automated approach for evaluating the energy performance and detecting faults in heating, ventilation, and air-conditioning (HVAC) systems. The study addressed the significant energy losses associated with poor maintenance and undetected equipment degradation by utilizing a systems-theory modeling approach integrated with continuous sensor data and analytical fault detection techniques. Rather than

relying solely on building automation systems, the proposed method continuously compares measured operating conditions with mathematical models of expected system behavior to identify abnormalities and performance deterioration. The findings indicated that automated performance assessment and fault detection can improve maintenance practices and achieve energy savings of up to 25% by identifying inefficiencies before they develop into major system failures.

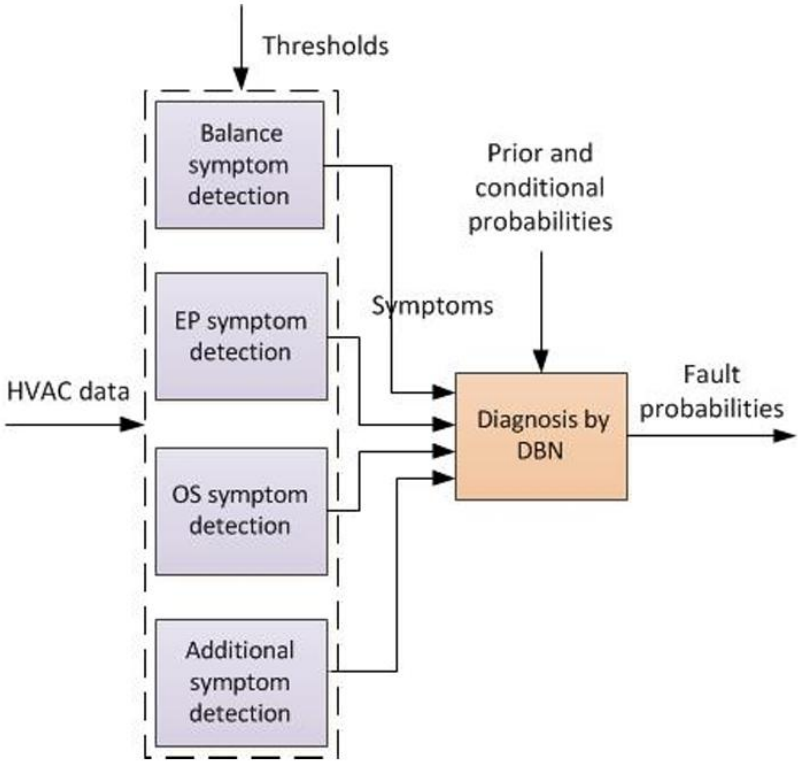


Figure 9

Automated Energy Performance Diagnosis Architecture

Figure 9 presents the architecture for automated energy performance fault detection and diagnosis proposed by Taal (2021). The framework separates the detection of system symptoms from the diagnosis of possible faults. Energy performance, operational state, balance, and additional symptoms are evaluated during the detection stage, while the diagnostic

stage processes the detected symptoms to identify possible model, component, and control faults. The architecture provides a systematic framework for transforming HVAC system information and measured performance data into automated diagnostic results.

DS18B20 Temperature Sensor for Air Temperature Measurement

Elyounsi and Kalashnikov (2021) evaluated the suitability of the DS18B20 digital temperature sensor for use in an accurate air temperature distribution measurement network by comparing its performance with an industrial-grade Pt100 resistance temperature detector (RTD). The study investigated the effects of sensor response time, unrestricted air movement, and networking capability on air temperature measurements. Experimental results demonstrated that, after minimizing external influences, the DS18B20 exhibited excellent agreement with the Pt100 sensor, with an average measurement bias of only 0.05 °C, confirming its capability for accurate quasi-stationary air temperature measurements. Furthermore, the authors highlighted that the DS18B20 allows multiple sensors to communicate through a single three-wire bus, making it a practical and cost-effective solution for temperature measurement networks requiring several sensing points. These characteristics demonstrate the suitability of the DS18B20 for indoor environmental measurement applications and multi-point temperature monitoring systems.

SCT-013-000 Current Sensor and ZMPT101B Voltage Sensor for Electrical Parameter Measurement

Nugraha et al. (2024) developed an Internet of Things (IoT)-based real-time electricity consumption monitoring system that utilized the SCT-013-000 current sensor and the ZMPT101B voltage sensor to measure electrical parameters. The SCT-013-000 sensor measured alternating current based on the current transformer principle, providing galvanic isolation between the primary and secondary circuits, while the ZMPT101B sensor measured

AC voltage with high precision and stability for real-time monitoring applications. The measured voltage and current values were transmitted to a microcontroller for processing and visualization through an IoT platform, enabling remote monitoring of electrical energy consumption. Experimental evaluation demonstrated that the SCT-013-000 achieved an average current measurement accuracy of 93.94%, whereas the ZMPT101B obtained an average voltage measurement accuracy of 98.50% when compared with standard measuring instruments. These results demonstrated that both sensors provide reliable and cost-effective solutions for real-time electrical parameter measurement in IoT-based energy monitoring systems.

FS3000 Air Velocity Sensor for Indoor Airflow Measurement

Ospina-Rojas et al. (2025) developed an open-source Internet of Things (IoT)-based three-dimensional anemometer for indoor air quality monitoring using the FS3000 MEMS airflow sensor integrated with an ESP32 microcontroller. The FS3000 operates based on the hot-wire thermal principle, wherein an internal heating element is cooled by the surrounding airflow, and the resulting temperature variation is detected by a thermopile to determine air velocity. The sensor communicates through the I²C interface with a 12-bit resolution, has a measurement range of 0–7.23 m/s, an accuracy of ±5%, and is specifically designed for indoor airflow monitoring applications. Experimental validation demonstrated that the proposed anemometer successfully detected variations in indoor air velocity with a deviation of 0.25 m/s, confirming its capability for accurate airflow measurement in enclosed environments. The researchers further emphasized that the compact design, low power consumption, and compatibility with ESP32-based IoT systems make the FS3000 suitable for ventilation monitoring, indoor environmental assessment, and air-conditioning performance evaluation.

Indoor Lighting Standards for Workplaces and Educational Facilities

The European Standard EN 12464-1:2021 establishes the lighting requirements for indoor workplaces to ensure visual comfort, safety, and effective task performance. The standard specifies lighting criteria such as the minimum maintained illuminance (Em), lighting uniformity (Uo), unified glare rating (UGR), and color rendering index (Ra) for various indoor environments such as classrooms, offices, laboratories, libraries, and meeting rooms. It recommends that lighting design be based on the visual tasks performed within each space while maintaining adequate illumination quality and occupant comfort. For educational facilities, the standard recommends maintained illuminance levels appropriate for classrooms, lecture halls, computer laboratories, libraries, and other learning spaces, providing a consistent basis for evaluating indoor lighting performance.

Ref.	Area / Activity	Em (lx)	UGRL	Uo	Ra	Notes
5.2.1	Filing, copying, circulation	300	19	0,40	80	
5.2.2	Writing, reading, data processing	500	19	0,60	80	For CAD work see 5.2.3
5.2.3	CAD workstations	500	19	0,60	80	Special attention to screen
5.2.4	Meeting rooms	500	19	0,60	80	Lighting should be dimmable
5.2.5	Reception desk	300	22	0,40	80	For reading/writing at counter: 500 lx
5.2.6	Archives	200	25	0,40	80	

Figure 10

Maintained Illuminance Requirements for Office Spaces

Figure 10 presents the maintained illuminance requirements for selected office spaces based on the lighting specifications of EN 12464-1. The specified illuminance levels vary according to the type of office activity and the visual demands of tasks performed within the

area. These requirements provide quantitative criteria for determining appropriate illumination levels for office work, meetings, filing, and other workplace activities.

Ref.	Area / Activity	Em (lx)	UGRL	Uo	Ra	Notes
5.5.1	Kindergarten / nursery	300	19	0,40	80	Floor level as reference surface
5.5.2	Classrooms	300	19	0,60	80	Dimmable lighting
5.5.3	Adult / evening education rooms	500	19	0,60	80	Special visual comfort requirements
5.5.4	Lecture halls / auditoriums	500	19	0,60	80	Dimmable lighting
5.5.5	Blackboard / whiteboard	500	19	0,70	80	Avoid reflections · vertical surface
5.5.6	Demonstration / presentation board	500	19	0,70	80	Vertical surface
5.5.7	Art / craft workshops	500	19	0,60	80	
5.5.8	Music practice rooms	300	19	0,60	80	
5.5.9	Computer labs	300	19	0,60	80	Screen reflection control
5.5.10	Language labs	300	19	0,60	80	
5.5.11	Teacher preparation rooms	300	22	0,60	80	
5.5.12	Gymnasiums (general use)	300	22	0,40	80	500 lx for sports
5.5.13	Libraries — shelves	200	19	0,40	80	Vertical lighting
5.5.14	Libraries — reading area	500	19	0,60	80	

Figure 11

Maintained Illuminance Requirements for Educational Spaces

Figure 11 presents the maintained illuminance requirements for selected educational spaces based on the lighting specifications of EN 12464-1. The required illuminance levels vary according to the function and visual tasks performed in classrooms, lecture rooms, laboratories, libraries, and other learning areas. These values provide quantitative criteria for determining the appropriate illumination levels required to support visual comfort and task performance in educational environments.

Thermal Environmental Conditions for Human Occupancy (ASHRAE Standard 55-2023)

The American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) developed ASHRAE Standard 55-2023: Thermal Environmental Conditions for Human Occupancy to establish the thermal environmental conditions that provide acceptable indoor thermal comfort for building occupants. The standard specifies the environmental and personal factors that influence thermal comfort, including air temperature, humidity, air speed, mean radiant temperature, clothing insulation, and metabolic rate. It provides standardized methods for evaluating thermal conditions in occupied buildings and defines acceptable comfort zones intended to satisfy the majority of occupants. The standard also serves as a reference for assessing existing indoor environments and verifying compliance with thermal comfort requirements, making it widely recognized in the design, operation, and evaluation of heating, ventilation, and air-conditioning (HVAC) systems.

Continuous Indoor Environmental Quality (IEQ) Measuring Systems

Parkinson et al. (2019) investigated the performance of a continuous Indoor Environmental Quality (IEQ) measuring system designed to assess building conditions through low-cost environmental sensors. The study evaluated the capability of continuous measuring systems in measuring indoor environmental parameters and classifying thermal comfort conditions over extended periods. The authors emphasized that continuous measuring provides a more representative assessment of building performance than traditional single-point measurements because it captures temporal and spatial variations in indoor environmental conditions. The study further demonstrated that low-cost sensor networks can provide sufficient accuracy for building diagnostics, compliance assessments, and long-term facility performance evaluation.

	ISO 7726:2001	ASHRAE 55-2017
Air Temperature		
Range:	10 to 40°C	10 to 40°C
Accuracy:	Required: ± 0.5°C Desirable: ± 0.2°C	± 0.2°C
Mean Radiant Temperature		
Range:	10 to 40°C	10 to 40°C
Accuracy:	Required: ± 2°C Desirable: ± 0.2°C	± 1°C
Humidity		
Range:	0.5 kPa to 3.0 kPa	25 to 95%
Accuracy:	± 0.15 kPa	± 5%
Air Velocity		
Range:	0.05 m/s to 1.00 m/s	0.05 m/s to 2.00 m/s
Accuracy:	Required: ±(0.05 + 0.05*va*) m/s Desirable: ±(0.02 + 0.07*va*) m/s	±0.05 m/s

Figure 12

Instrumentation Requirements for Thermal Comfort Assessment

Figure 12 summarizes the instrumentation requirements for thermal comfort assessment based on recognized standards. It presents the recommended measurement ranges and accuracy for key indoor environmental parameters, providing guidance for selecting appropriate measuring instruments used in indoor environmental standards.

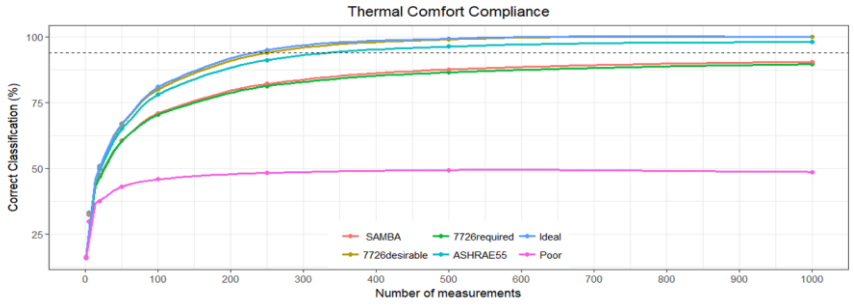


Figure 13

Classification Performance of Continuous Measuring Systems

Figure 13 illustrates the classification performance of continuous measuring systems under different measurement conditions. The figure shows that increasing the amount of continuously collected environmental data improves the reliability of indoor environmental assessment and supports more representative performance evaluations over time.

Energy Efficiency and Digitalisation

The International Energy Agency (IEA, 2019) emphasized that digital technologies are transforming energy efficiency by enabling greater control, optimization, and analysis of energy-consuming systems through real-time data acquisition. It explains that digitalization integrates sensors, wireless communication, smart meters, and data analytics to continuously collect and process information, allowing buildings and facilities to improve operational performance and reduce unnecessary energy consumption. Furthermore, it highlights that digital technologies support the development of smart buildings and intelligent energy management systems by enabling connected and data-driven approaches to improving end-use efficiency and overall energy system performance.

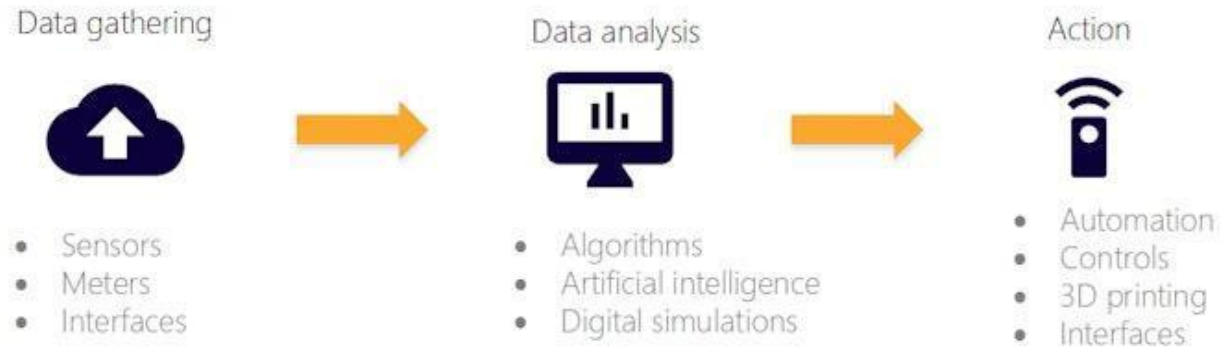


Figure 14

Digitalisation Process for Energy Efficiency

Figure 14 illustrates how digital technologies improve energy efficiency through the integration of data gathering, data analysis, and action. According to the International Energy Agency (2019), sensors and smart meters collect energy-use data and other relevant information, while data analysis technologies process the collected data into meaningful insights. These insights support either automated systems or human decision-making to optimize energy use and improve overall energy efficiency. The integration of data gathering, data analysis, and action plays a vital role in enhancing energy performance by enabling informed and timely decisions based on accurate and continuously collected data.

Implementing Rules and Regulations of the Energy Efficiency and Conservation Act

The Department of Energy (DOE, 2019) issued Department Circular No. DC2019-11-0014, which contains the Implementing Rules and Regulations (IRR) of Republic Act No. 11285, also known as the Energy Efficiency and Conservation (EE&C) Act. The circular establishes a national framework for promoting energy efficiency and conservation by requiring government agencies, educational institutions, commercial establishments, and other designated facilities to implement energy management practices, conduct energy audits, measure energy consumption, and adopt energy-efficient technologies. It also emphasizes the importance of continuous measuring, evaluation, reporting, and compliance as essential components of sustainable facility and energy management.

Synthesis of the Review of Related Literature

The reviewed literature and related studies collectively establish the technical foundation for the development of the proposed Smart Facility Measuring System for Indoor Lighting and Air-Conditioning and Their Energy Efficiency. Previous studies have demonstrated that Internet of Things (IoT)-based measuring systems, wireless sensor networks, and ESP32

microcontrollers provide reliable and cost-effective platforms for real-time environmental measuring and facility management. These technologies have been successfully applied in acquiring sensor measurements, transmitting data wirelessly, and supporting centralized measuring through web-based and mobile applications.

Similarly, studies on indoor lighting assessment have emphasized the importance of maintaining appropriate illumination levels in accordance with established standards such as EN 12464-1 to improve occupant comfort, productivity, and energy conservation. In addition, the literature highlights that indoor lighting should be evaluated through multi-point illumination measurements using representative measurement grids because illumination is not uniformly distributed across a room. The standardized procedures developed by the National Renewable Energy Laboratory (NREL) provide a technical basis for determining the number and placement of measurement points according to room dimensions and required measurement coverage, thereby improving the accuracy and representativeness of indoor lighting assessment.

Likewise, research related to heating, ventilation, and air-conditioning (HVAC) systems has shown that measuring electrical consumption, temperature, and airflow parameters enables the evaluation of cooling performance and energy efficiency while supporting preventive maintenance and reducing operational costs.

The reviewed literature also highlights the increasing adoption of centralized databases, cloud-based measuring platforms, and automated assessment algorithms for processing sensor data and generating meaningful information for facility managers. However, most existing systems focus on either indoor lighting or HVAC measuring independently. Furthermore, many studies emphasize real-time measuring but provide limited integration of lighting assessment, air-conditioning energy efficiency evaluation, wireless sensor deployment, automated recommendation generation, and multi-platform accessibility within a single measuring system.

To address these limitations, the present study will develop an integrated Smart Facility Measuring System capable of simultaneously measuring indoor lighting and air-conditioning operating conditions using ESP32-based wireless sensor nodes. The developed system will acquire, process, and analyze illumination, electrical, thermal, and airflow parameters through a centralized database while providing real-time visualization using a web-based dashboard and a Flutter mobile application. The lighting measurement component will adopt a multi-point sensor deployment strategy based on room dimensions and representative measurement coverage, following established lighting measurement procedures. In addition, the system will automatically generate assessments and recommendations based on recognized engineering standards and methodologies, thereby providing a comprehensive tool for improving facility operation, supporting preventive maintenance, and promoting energy-efficient building management.

Chapter 3

Methodology

This study employs a Design and Development Research approach in designing, developing, and evaluating a Smart Facility Measuring and Assessment System for indoor lighting assessment and air-conditioning energy efficiency analysis. The methodology is guided by the works of Zafar et al. (2018) on IoT-based environmental measuring, the National Renewable Energy Laboratory (NREL) lighting measurement procedures, Wang and Burnett (2001) on Internet-based HVAC measuring, Cline (2010) on air-conditioning cooling capacity testing, and Taal (2021) on automated HVAC energy performance assessment. These studies provide the foundation for developing a portable measuring system capable of acquiring, analyzing, and assessing indoor environmental and energy-related parameters.

The proposed system utilizes ESP32-based measuring modules equipped with illumination, electrical, thermal, and airflow sensors to acquire indoor environmental and energy-related measurements. The collected data are transmitted wirelessly via Wi-Fi to the developed Flutter-based mobile and desktop application for real-time measuring, visualization, assessment, and recommendation generation. For indoor lighting assessment, the system adopts the National Renewable Energy Laboratory (NREL) multi-point illumination measurement procedure and evaluates the computed average illumination level against the requirements of EN 12464-1 to determine lighting adequacy and generate appropriate lighting and energy conservation recommendations. For air-conditioning assessment, the system acquires electrical, thermal, and airflow parameters to estimate the cooling capacity and energy efficiency of the air-conditioning unit using the air-side methodology presented by Cline (2010). The estimated energy efficiency is compared with the manufacturer's rated performance, while

the measured indoor thermal conditions are evaluated based on the recommendations of ANSI/ASHRAE Standard 55-2023. Furthermore, the airflow measurement principles presented by Rønneseth et al. (2018) and the automated energy performance assessment concepts proposed by Taal (2021) provide the basis for the system's automated assessment and recommendation generation.

Unlike previous studies that focused only on lighting assessment or HVAC measuring, the proposed system integrates indoor lighting assessment, air-conditioning energy efficiency analysis, automated evaluation, and recommendation generation into a single portable measuring platform. By combining these functions in a centralized Flutter application, the system provides facility managers with a practical decision-support tool for evaluating indoor environmental conditions, promoting energy conservation, and supporting preventive maintenance.

Locale of the Study

The study will be conducted at the College of Engineering Building of Central Philippine University (CPU), Iloilo City. The selected testing locations include one classroom, laboratory rooms, and faculty offices equipped with indoor lighting systems and split-type air-conditioning units. These facilities provide representative indoor environments suitable for evaluating the performance of the developed Smart Facility Measuring System under different lighting conditions, room sizes, occupancy levels, and air-conditioning configurations. The selected locations will serve as the testing sites for evaluating the functionality, reliability, and accuracy of the developed prototype. Table 1 presents the selected testing locations and their corresponding purpose in the study.

Table 1

Selected Testing Locations

Location	Purpose in the Study
Classroom (EN 106)	Evaluate the system under typical teaching and learning conditions, where lighting quality, thermal comfort, and air-conditioning energy efficiency are essential for occupant comfort and productivity.
Laboratory Rooms (EnA 202 & Drawing Room)	Evaluate the system under varying lighting conditions, equipment usage, occupancy levels, and air-conditioning operation commonly experienced during laboratory activities.
Faculty Offices (Dean's Office & EE Office)	Assess the system in office environments with relatively stable occupancy, lighting conditions, and air-conditioning operation to determine its applicability in administrative spaces.

System Development and Preparation

The development of the proposed Smart Facility Measuring System for Indoor Lighting and Air-Conditioning and Their Energy Efficiency will involve the systematic planning, design, fabrication, integration, and testing of both the hardware and software components. The system will be developed as a portable and wireless measuring platform capable of assessing indoor lighting conditions and evaluating the operational energy efficiency of split-type air-conditioning systems through the acquisition of illumination, electrical, thermal, and airflow parameters. The collected measurements will be processed to perform lighting assessment, estimate cooling capacity and energy efficiency, evaluate indoor thermal conditions, and generate recommendations that will support energy conservation and facility maintenance.

The overall system architecture will be developed by integrating the Internet of Things (IoT) measuring concepts presented by Zafar et al. (2018) and Yang et al. (2025), the ESP32-based wireless sensor network architecture proposed by Khan et al. (2022), the lighting measurement procedures recommended by the National Renewable Energy Laboratory (NREL), the HVAC measuring architecture presented by Wang and Burnett (2001), the airflow measurement principles discussed by Rønneseth et al. (2018), the cooling capacity evaluation procedure proposed by Cline (2010), and the automated HVAC energy assessment concepts developed by Taal (2021). These standards and methodologies will serve as the technical foundation for developing an integrated Smart Facility Measuring System capable of acquiring real-time indoor environmental and energy-related data, performing automated assessments, and presenting the processed information through a centralized measuring platform.

The system development process will be presented through the overall system architecture, hardware design, software design, and operational workflow of the proposed prototype. These components will collectively describe the implementation of the Smart Facility Measuring System, from sensor deployment and wireless data acquisition to real-time measuring, data processing, assessment, and recommendation generation.

Hardware Development

The proposed Smart Facility Measuring and Assessment System for Indoor Lighting and Air-Conditioning Performance will consist of two ESP32-based measuring modules, a Wi-Fi communication network, and a Flutter-based mobile and desktop application that enables the real-time acquisition, transmission, visualization, assessment, and storage of indoor environmental and energy-related measurements. The Indoor Lighting Measuring Module integrates BH1750 illumination sensors, while the Air-Conditioning Measuring Module incorporates voltage, current, temperature, and air velocity sensors to acquire lighting, electrical, thermal, and airflow parameters from the measured indoor facility. The acquired measurements

are transmitted wirelessly via the built-in Wi-Fi capability of the ESP32 modules to the developed Flutter application for centralized measuring and assessment.

The Flutter-based mobile and desktop application serves as the central measuring platform of the proposed system. It receives and stores the acquired measurements, computes the average illumination level, estimates the cooling capacity and energy efficiency of the air-conditioning unit, and evaluates indoor thermal conditions. The application presents the collected data through a real-time measuring dashboard and automatically performs indoor lighting assessment based on EN 12464-1, air-conditioning energy efficiency evaluation using the adopted air-side methodology, and classroom thermal condition assessment based on ANSI/ASHRAE Standard 55-2023. Based on these assessments, the system automatically generates maintenance and energy conservation recommendations to support facility management, preventive maintenance, and energy-efficient operation.

Figure 15 illustrates the overall architecture of the proposed Smart Facility Measuring and Assessment System, showing the interaction between the Indoor Lighting Measuring Module, Air-Conditioning Measuring Module, Wi-Fi communication network, Flutter-based mobile and desktop application, and the assessment and recommendation module.

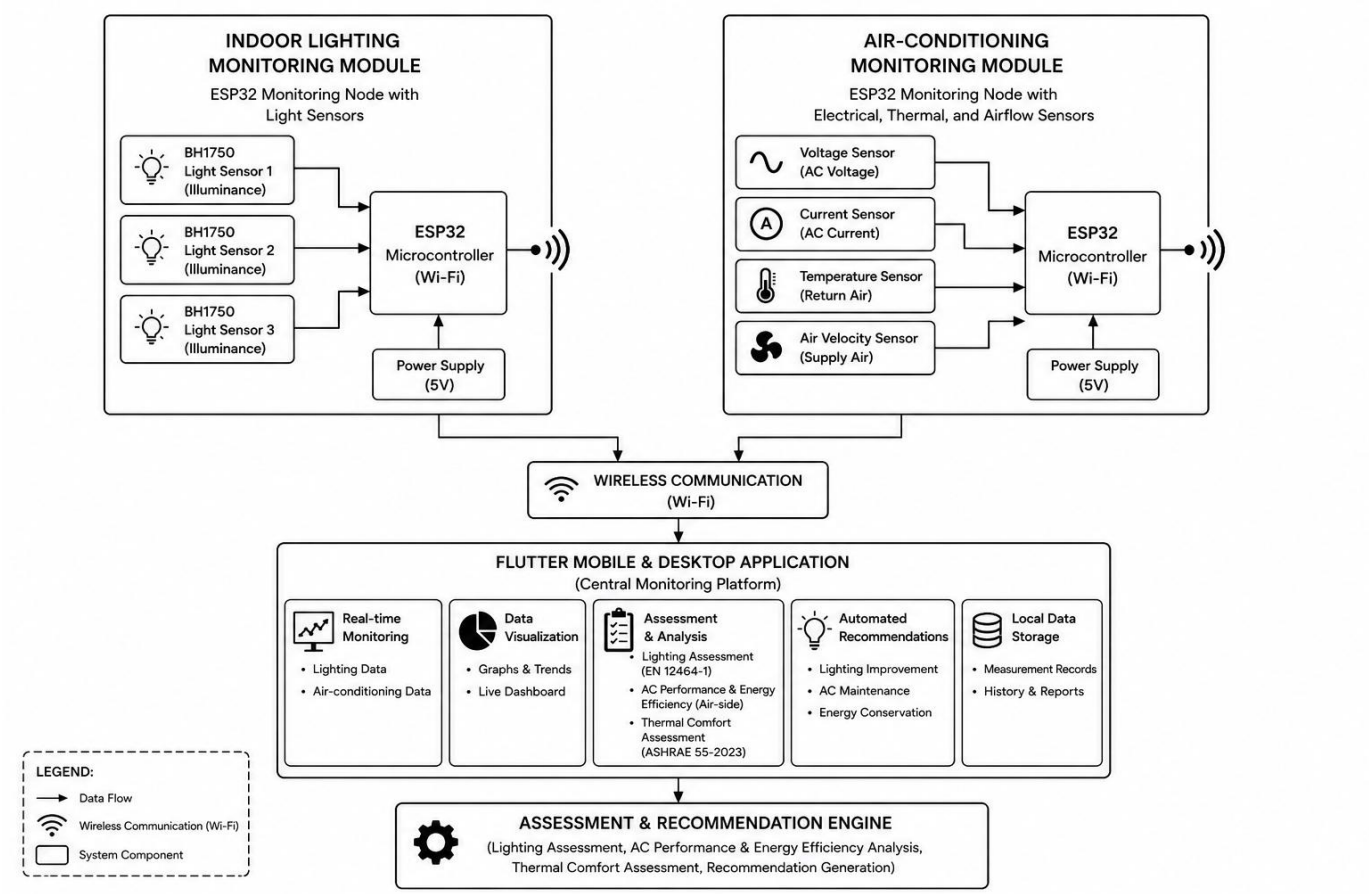


Figure 15.

Overall System Architecture

Figure 15 illustrates the overall architecture of the proposed Smart Facility Measuring and Assessment System. The system consists of two ESP32-based measuring modules: the Indoor Lighting Measuring Module and the Air-Conditioning Measuring Module, which acquire illumination, electrical, thermal, and airflow measurements from the measured indoor facility through integrated sensors. The acquired measurements are transmitted wirelessly via the built-in Wi-Fi capability of the ESP32 modules to the developed Flutter-based mobile and desktop application. The application serves as the central measuring platform, where the collected data are visualized, processed, stored, and assessed in real time. It computes the average

illumination level, evaluates indoor lighting conditions based on EN 12464-1, estimates the cooling capacity and energy efficiency of the air-conditioning system, and assesses classroom thermal conditions in accordance with ANSI/ASHRAE Standard 55-2023. Based on these assessments, the application automatically generates maintenance and energy conservation recommendations to support facility measuring, data-driven decision-making, and preventive maintenance.

Software Development

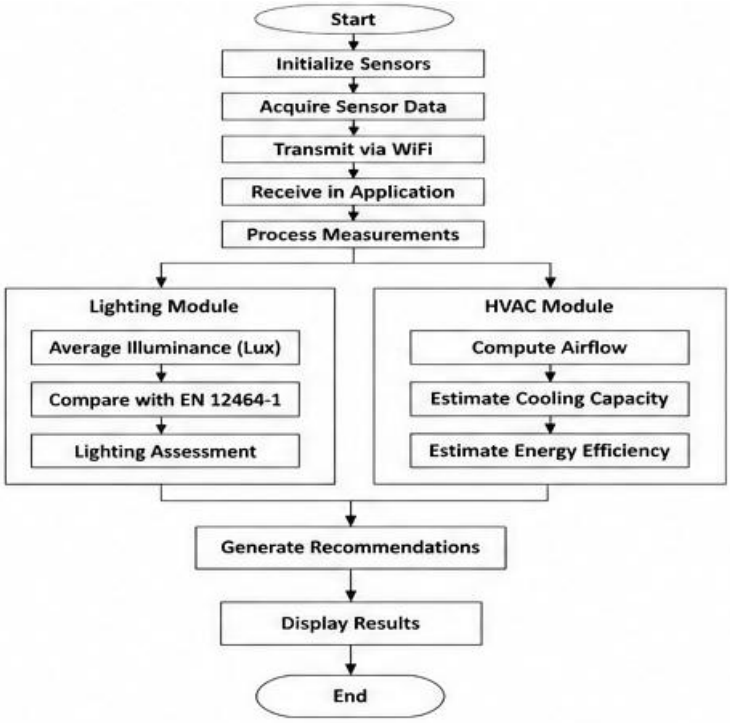


Figure 16.

Software Processing Flow of the Smart Facility Measuring System

Figure 16 illustrates the software architecture of the proposed Smart Facility Measuring and Assessment System. The system utilizes a Flutter-based application,

which serves as the user interface for both desktop and mobile platforms. The Flutter application receives real-time sensor data transmitted by the ESP32 module, processes and stores the collected measurements, and performs indoor lighting assessment and air-conditioning energy efficiency analysis. The application computes the average illumination level and compares it with the EN 12464-1 lighting requirements, while the air-conditioning module estimates the cooling capacity and energy efficiency, compares the estimated energy efficiency with the manufacturer's rated performance, and evaluates the classroom thermal conditions based on ANSI/ASHRAE Standard 55-2023. Based on these assessments, the application automatically generates maintenance and energy conservation recommendations, which are displayed through the dashboard for real-time measuring and decision-making.

Assessment and Recommendation Algorithm

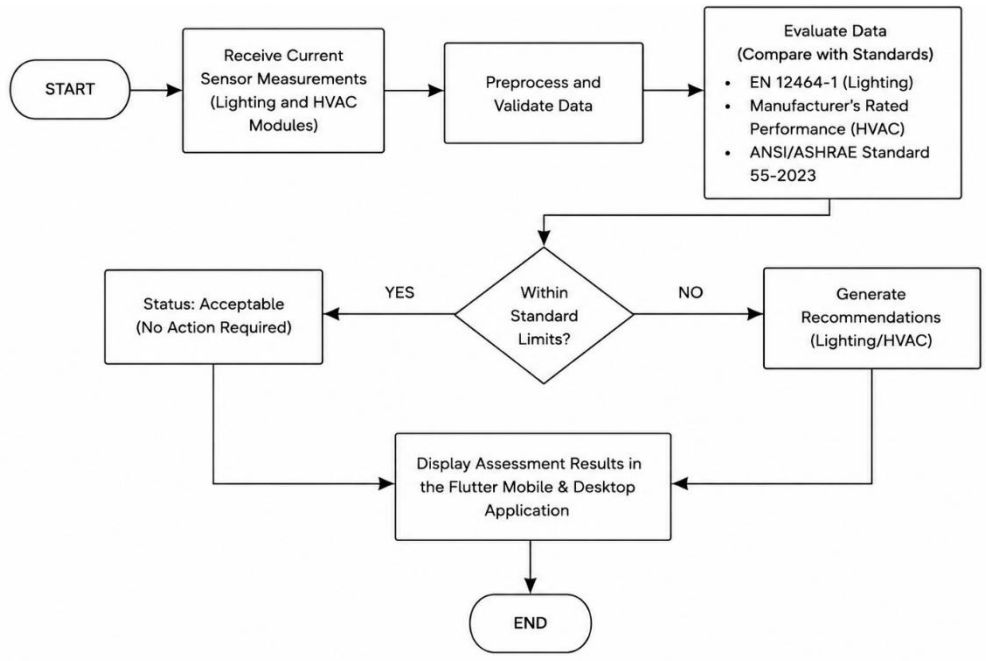


Figure 17.

Assessment and Recommendation Algorithm

Figure 17 illustrates the Assessment and Recommendation Algorithm of the proposed Smart Facility Measuring System. The algorithm begins by receiving the current sensor measurements from the indoor lighting and air-conditioning measuring modules, followed by data preprocessing and validation. The validated measurements are then evaluated against the prescribed standards, including EN 12464-1 for indoor lighting, the manufacturer's rated performance for the air-conditioning unit, and ANSI/ASHRAE Standard 55-2023 for thermal comfort. If the measured parameters comply with the established standards, the system indicates that no corrective action is required. Otherwise, appropriate recommendations are automatically generated. The assessment results and recommendations are then displayed through the developed Flutter-based mobile and desktop application for real-time facility measuring and decision support.

Operational Workflow

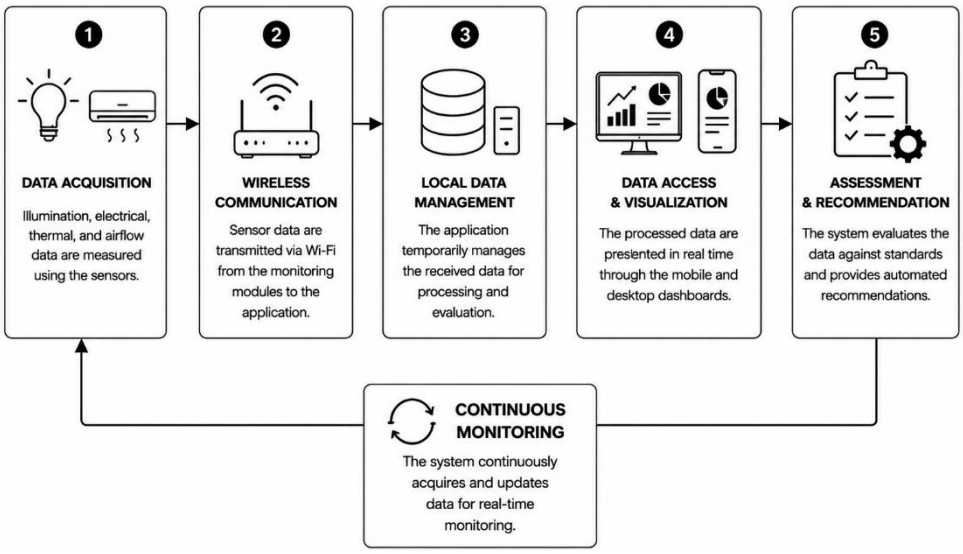


Figure 18.

Operational Workflow

Figure 18 illustrates the operational workflow of the proposed Smart Facility Measuring and Assessment System. The workflow begins with the acquisition of indoor lighting and air-conditioning data through the respective measuring modules. The acquired measurements are then transmitted wirelessly via Wi-Fi to the developed Flutter-based mobile and desktop application, where the data are temporarily processed and evaluated against the prescribed standards. Based on the evaluation results, the system automatically generates facility assessment results and appropriate recommendations, which are displayed to the user to support real-time measuring and informed facility management. The process operates continuously, enabling ongoing assessment of indoor environmental conditions.

The hardware assembly will consist of integrating the ESP32 microcontrollers with the required lighting, electrical, thermal, and airflow sensors. Each measuring module will be assembled according to the developed hardware design to ensure accurate sensor measurements and reliable wireless communication. Proper power distribution, wiring, and sensor placement will be implemented to support stable system operation.

System Integration

The system integration will combine the hardware and software components into a unified Smart Facility Measuring System. The integrated prototype will undergo functional verification to ensure that the measuring modules, wireless communication, and measuring applications operate as a complete system. The integrated platform will provide continuous measuring, automated assessment, and recommendation generation for indoor lighting and air-conditioning energy efficiency.

Hardware Design

The hardware design of the proposed Smart Facility Measuring and Assessment System for Indoor Lighting and Air-Conditioning Performance consists of the components required for measuring indoor lighting conditions and air-conditioning operating parameters. The system integrates two ESP32-based measuring modules equipped with illumination, electrical, thermal, and airflow sensors to acquire real-time measurements from the measured indoor facility. The Indoor Lighting Measuring Module utilizes BH1750 light sensors to measure illumination levels, while the Air-Conditioning Measuring Module incorporates voltage, current, temperature, and air velocity sensors to measure the operating parameters of the air-conditioning unit. The acquired measurements are transmitted wirelessly via the built-in Wi-Fi capability of the ESP32 modules to the developed Flutter-based mobile and desktop application, where the collected data are measured, processed, stored, and assessed in real time.

Determination of Lux Sensor Deployment

The number of BH1750 lux sensors to be deployed shall be determined based on the room dimensions and the required measurement coverage. The room will be divided into representative measurement areas following the multi-point measurement approach recommended by NREL and EN 12464-1. Sensor modules will be positioned at predetermined locations on the working plane (approximately 0.75 m above the floor) to obtain representative illumination measurements. Additional sensor modules will be added for larger rooms to maintain adequate spatial coverage and improve the accuracy of the computed average illuminance.

Table 2

Proposed Number of BH1750 Lux Sensors Based on Room Size and Measurement Coverage

Room Size	Proposed Number of BH1750 Sensors
Small (up to 20m^2)	3 sensors
Medium (20-50m^2)	5 sensors
Large (above 50m^2)	7 or more sensors

Indoor Lighting Measuring Module

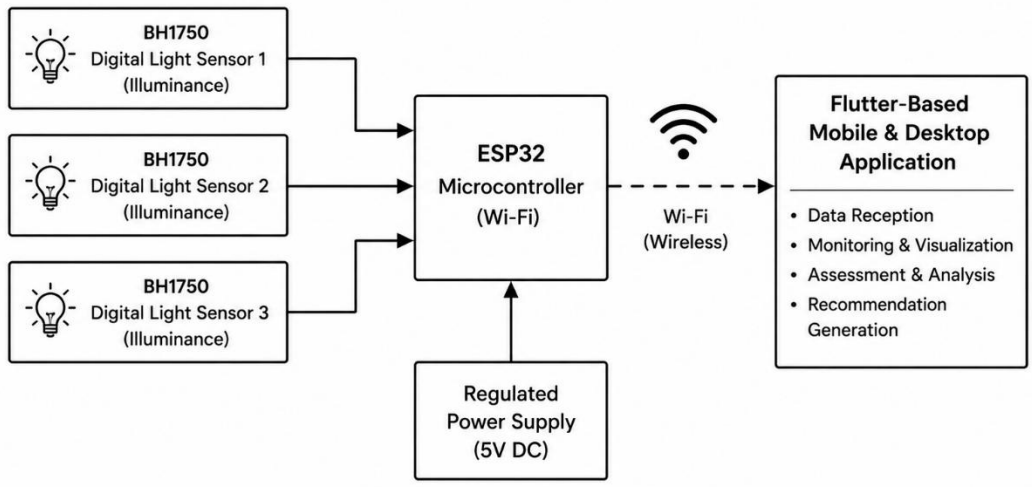


Figure 19.

Indoor Lighting Measuring Module

Figure 19 illustrates the hardware configuration of the proposed Indoor Lighting Measuring Module. The module consists of an ESP32 microcontroller, three BH1750 digital light sensors, and a regulated 5 V DC power supply. The BH1750 sensors measure the illumination level in lux (lx) at predetermined measurement nodes, while the ESP32 acquires the measured

data and transmits the collected illumination measurements wirelessly through its built-in Wi-Fi capability to the developed Flutter-based mobile and desktop application. The application receives, stores, and processes the acquired measurements to support real-time indoor lighting measuring, assessment, and recommendation generation.

Illumination Measurement Setup

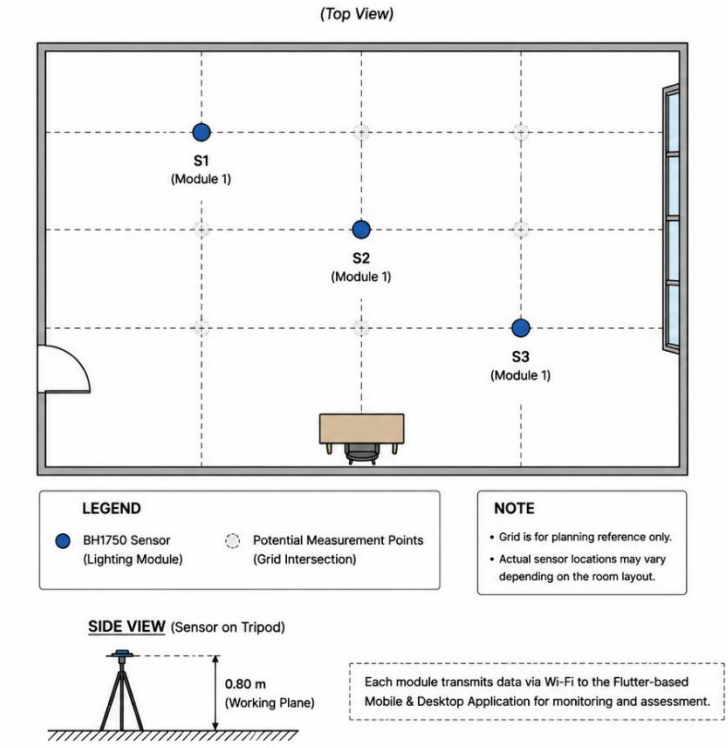


Figure 20.

Illumination Measurement Setup

Figure 20 illustrates the proposed illumination measurement setup of the Indoor Lighting Measuring Module. The illumination sensors are positioned at predetermined measurement nodes with approximately uniform spacing over the occupied working area to obtain representative illuminance measurements. Each Lighting Measuring Module

consists of one ESP32 microcontroller connected to three BH1750 digital light sensors. For larger rooms requiring additional measurement points, multiple Lighting Measuring Modules may be deployed while maintaining uniform sensor distribution. Each sensor is mounted on a tripod at a height of 0.80 m above the finished floor, with the sensing surface facing upward, following the adopted NREL multi-point illumination measurement procedure and the working-plane measurement concept of EN 12464-1.

Air-Conditioning Measurement Module

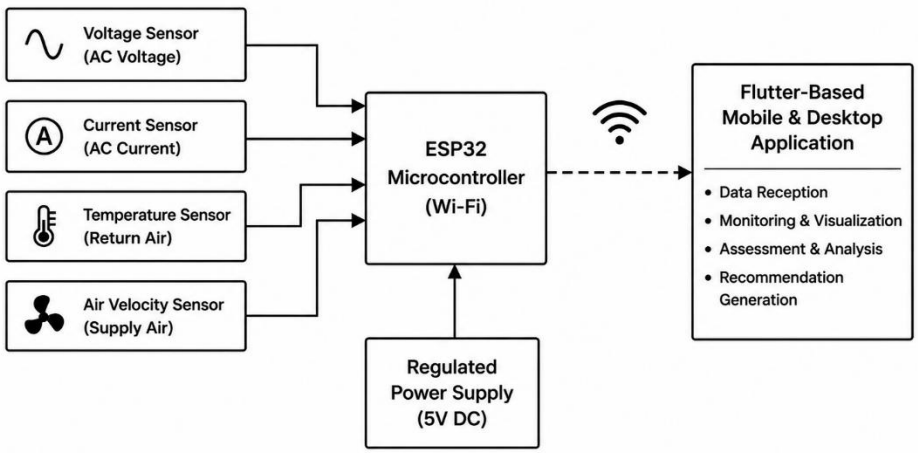


Figure 21.

Air-Conditioning Measuring Module

Figure 21 illustrates the hardware configuration of the proposed Air-Conditioning Measuring Module. The module consists of an ESP32 microcontroller integrated with voltage, current, temperature, and air velocity sensors. The ESP32 acquires the measured electrical, thermal, and airflow parameters and transmits the collected data

wirelessly through its built-in Wi-Fi capability to the developed Flutter-based mobile and desktop application for real-time measuring, visualization, and assessment.

The Air-Conditioning Measuring Module acquires the electrical and environmental parameters required for evaluating the operating performance of the air-conditioning system. The voltage and current sensors measure the electrical input of the air-conditioning unit, while the temperature and air velocity sensors acquire the thermal and airflow data required for estimating the cooling capacity and energy efficiency using the adopted air-side evaluation methodology. The collected measurements are transmitted by the ESP32 to the Flutter application, where the cooling capacity, estimated energy efficiency, and classroom thermal conditions are evaluated. Based on the assessment results, the application automatically generates maintenance and energy conservation recommendations.

Air-Conditioning Measurement Setup

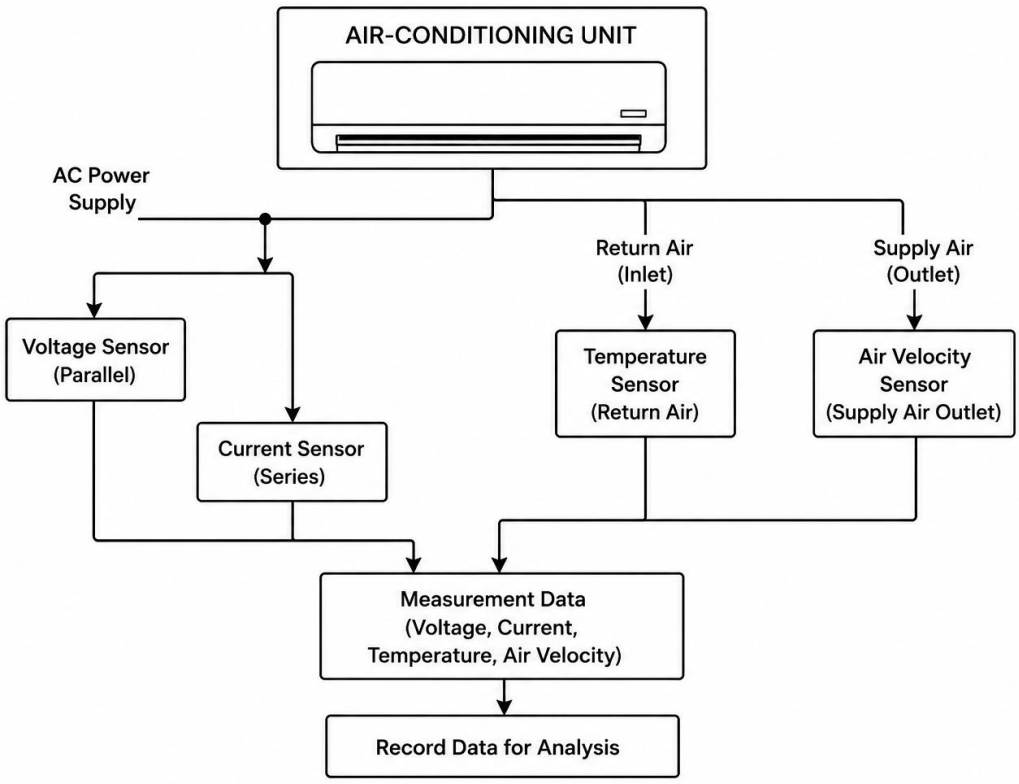


Figure 22.

Air-Conditioning Measurement Setup

Figure 22 illustrates the measurement setup used to evaluate the operating performance and energy efficiency of the air-conditioning unit. The voltage sensor is connected in parallel with the AC power supply to measure the operating voltage, while the current sensor is connected in series with the live conductor to measure the current drawn by the unit. The temperature sensor is positioned at the return-air inlet to measure the return-air temperature, and the air velocity sensor is placed at the supply-air outlet to measure the discharge air velocity. The measurements obtained from these sensors provide the electrical and thermal parameters required for evaluating the energy efficiency

of the air-conditioning system and serve as the input data for the proposed Smart Facility Measurement System.

Design Calculations

The developed Smart Facility Measuring and Assessment System performs a series of engineering calculations to transform the acquired sensor measurements into meaningful assessment parameters. These calculations enable the system to evaluate indoor lighting conditions and the energy efficiency of air-conditioning systems by processing the measured illumination, electrical, thermal, and airflow data. The resulting values serve as the basis for the automated assessment and recommendation features of the developed application.

Average Illumination Level

The average illumination level shall be computed by averaging the illumination measurements obtained from all predetermined measurement points within the measured facility. This computation follows the multi-point lighting assessment procedure recommended by the National Renewable Energy Laboratory (NREL) and is used to evaluate compliance with the illumination requirements specified in EN 12464-1.

$$\bar{E} = \frac{\sum_{i=1}^{n} E_i}{n}$$

P = Electrical input power (W)

V = Measured voltage (V)

I = Measured current (A)

Electric Input Power

The electrical input power of the measured air-conditioning system shall be computed from the measured voltage and current. This equation is a fundamental electrical engineering relationship and is used as part of the air-conditioning energy efficiency analysis adopted by Cline (2010).

$$P = VI$$

Where:

P = Electrical input power (W)

V = Measured voltage (V)

I = Measured current (A)

Airflow Rate

The airflow rate shall be determined from the measured air velocity and the cross-sectional area of the air outlet. This computation follows the airflow measurement methodology discussed by Rønneseth et al. (2018).

$$\dot{V} = vA$$

Where:

$\dot{V}$ = Airflow rate (m^3/s)

v = Measured air velocity (m/s)

A = Air outlet cross-sectional area (m^2)

Cooling Capacity

The cooling capacity shall be estimated using the air-side evaluation methodology presented by Cline (2010), which utilizes airflow rate and the temperature difference between the room air and supply air.

$$Q = \rho c_p \dot{V} (T_r - T_s)$$

Energy Efficiency

The energy efficiency of the measured air-conditioning system shall be determined by comparing the estimated cooling capacity with the electrical input power. This analysis follows the HVAC energy performance assessment methodology presented by Cline (2010) and the automated performance evaluation concepts discussed by Taal (2021).

$$\eta = \frac{Q}{P}$$

Where:

η = Energy efficiency (W/W or COP)

Q = Cooling capacity (W)

P = Electrical input power (W)

Prototype Development Cost

Table 3.

Estimated Prototype Development Cost

Component	Quantity	Unit Cost (₱)	Total Cost (₱)
ESP32	4	450	1800
BH1750 Sensor	9	180	1620
Voltage Sensor	1	150	150
Current Sensor	1	180	180
Temperature Sensor	1	250	250
Air Velocity Sensor	1	2500	2500
Adjustable Tripod	3	250	750
Power Supply (5V Adapter)	4	200	800
Miscellaneous Materials	1 lot	1,200	1,200
Total Estimated Cost			**₱ 9,250**

Design Criteria

Functionality

The developed Smart Facility Measuring System shall be capable of performing all intended functions of the proposed design. Specifically, the system shall continuously acquire illumination, electrical, thermal, and airflow measurements using the integrated sensing devices; transmit the collected data wirelessly to the developed mobile and desktop application; estimate the cooling capacity and energy efficiency of the measured air-conditioning system, compare

the estimated energy efficiency with the manufacturer's rated performance, evaluate the classroom thermal conditions based on ANSI/ASHRAE Standard 55-2023, and automatically generate appropriate maintenance and energy conservation recommendations.

The functionality of the integrated system shall be verified through functional testing of both the hardware and software components. Functional testing shall confirm that the system successfully performs data acquisition, wireless data transmission, engineering calculations, automated assessment, data visualization, and recommendation generation in accordance with the adopted methodologies of Zafar et al. (2018), Yang et al. (2025), Wang and Burnett (2001), Cline (2010), and Taal (2021).

Accuracy

The developed Smart Facility Measuring System shall provide measurements that closely agree with those obtained from standard measuring instruments. Illumination measurements shall follow the procedures recommended by the National Renewable Energy Laboratory (NREL) and shall be validated using a commercial lux meter. Likewise, the voltage, current, temperature, and air velocity measurements shall be validated using corresponding reference instruments. The accuracy of the developed prototype shall be determined by comparing the acquired measurements with the reference measurements and computing the percentage error and measurement accuracy.

Reliability

The developed Smart Facility Measuring System shall demonstrate reliable operation by consistently performing its intended functions across different testing locations. Reliability shall be evaluated by deploying the developed system in five selected rooms and verifying the successful operation of its major functions, including data acquisition, wireless communication, engineering calculations, assessment generation, and recommendation generation. A testing

location shall be considered to have passed the reliability evaluation if all major system functions operate successfully throughout the assessment. The overall reliability of the developed system shall then be determined using the reliability equation based on the number of testing locations that successfully pass the evaluation.

Data Evaluation Procedure

The developed Smart Facility Measuring and Assessment System shall undergo functional, accuracy, and reliability testing prior to and during data collection to verify the performance of the developed prototype. Functional testing shall verify the proper operation of the integrated hardware and software components, following the IoT measuring and system integration methodologies of Zafar et al. (2018), Yang et al. (2025), and Wang and Burnett (2001). Accuracy testing shall compare the measurements obtained from the developed prototype with those obtained from standard reference instruments using three repeated measurements for each measured parameter. The average of the repeated measurements shall be used for validation to reduce random measurement error and improve measurement repeatability, consistent with standard engineering experimental practice and the measurement procedures adopted from the National Renewable Energy Laboratory (NREL), Rønneseth et al. (2018), and Cline (2010). Reliability testing shall be conducted by deploying the developed system in selected indoor facilities to verify its consistent operation under actual field conditions, following the continuous measuring principles discussed by Heinzerling et al. (2020) and the integrated IoT measuring approach demonstrated by Yang et al. (2025). The testing activities shall be documented using the succeeding Functional Testing Matrix, Accuracy Validation Sheet, and Reliability Testing Sheet, which serve as the basis for evaluating the functionality, accuracy, and reliability of the developed Smart Facility Measuring and Assessment System.

Functionality

Table 3 presents the Functionality Testing Matrix used to verify whether each function of the developed Smart Facility Measuring and Assessment System operates as intended by checking if each sensor and system feature successfully performs its designated operation. A function is considered Pass if it successfully displays the required reading or generates the intended output; otherwise, it is marked Fail.

Table 4.

Functionality Testing Matrix

System Function	Expected Output	Result (Pass/Fail)
Illumination Measurement	Displays illumination (lux)	
Voltage Measurement	Displays voltage (V)	
Current Measurement	Displays current (A)	
Temperature Measurement	Displays room and supply-air temperature	
Air Velocity Measurement	Displays air velocity (m/s)	
Wireless Data Transmission	Data successfully transmitted	
Lighting Assessment	Lighting condition displayed	
HVAC Energy Efficiency Analysis	Cooling capacity and energy efficiency computed	
Recommendation Generation	Appropriate recommendation displayed	

Accuracy

Accuracy testing will be conducted to determine the closeness of the measurements obtained by the proposed system to those obtained using calibrated reference instruments. The measured values from the developed system will be compared with reference measurements to determine the measurement error and evaluate the overall accuracy of the measuring modules. The following parameters will be evaluated Indoor illuminance (lux), Voltage (V), Current (A), Temperature (°C) , and Air velocity (m/s). Each parameter will be measured simultaneously using the proposed system and a corresponding calibrated reference instrument. The percentage error and measurement accuracy will be computed for comparison.

Table 5.

Accuracy testing Table for Illumination

Trial	BH1750 Sensor (lx)	(Lux Meter) (lx)	Percentage Error (%)
1			
2			
3			
Average			

Table 6.

Accuracy Testing Table for Voltage

Trial	Voltage Sensor (V)	(Digital Multi-meter) (V)	Percentage Error (%)
1			
2			
3			
Average			

Table 7.

Accuracy Testing Table for Current

Trial	Current Sensor (A)	Clamp Multi-meter (A)	Percentage Error (%)
1			
2			
3			
Average			

Table 8.

Accuracy Testing Table for Room Temperature

Trial	Temperature Sensor (°C)	Digital Thermometer(°C)	Percentage Error (%)
1			
2			
3			
Average			

Table 9.

Accuracy Testing Table for Supply-Air Temperature

Trial	Temperature Sensor (°C)	Digital Thermometer(°C)	Percentage Error (%)
1			
2			
3			
Average			

Table 10.

Accuracy Testing Table for Air Velocity

Trial	Air Velocity Sensor (m/s)	Anemometer (m/s)	Percentage Error (%)
1			
2			
3			
Average			

Reliability

Reliability testing will be performed to evaluate the consistency and stability of the proposed Smart Facility Measuring System during repeated operation. The system will be operated continuously under normal indoor conditions, and its ability to maintain stable sensor readings, uninterrupted wireless communication, and consistent data transmission will be assessed. The system will be operated continuously for a specified testing period. During the test, the following aspects will be measured: Continuous sensor operation, Wi-Fi connectivity, Database communication, Dashboard updates, Mobile application updates, and Assessment

generation Any communication failures, data loss, or abnormal system behaviour will be recorded and analysed.

Table 10 presents the Reliability Testing Sheet that will be used to evaluate the consistent operation of the proposed Smart Facility Measuring System under actual testing conditions. The operation of the Indoor Lighting Measuring Module, Air-Conditioning Measuring Module, wireless communication, assessment generation, and recommendation generation will be verified at each testing location. The results will be recorded to determine the overall reliability of the developed prototype

Table 11.

Reliability Testing Sheet

Testing Location	Lighting Module	HVAC Module	Wireless Communication	Assessment Generated	Recommendation Generated	Overall Result	Remarks
Dean's Office							
EE Office (EN 203)							
Drawing Room							
EnA 202							
EN 106							

Instrumentation

***Developed Smart Facility Measuring and Assessment System*.** The developed Smart Facility Measuring System shall serve as the primary instrument of the study. The system consists of an ESP32-based microcontroller integrated with a WiFi module, illumination sensor, voltage sensor, current sensor, temperature sensors, and an air velocity sensor. It is designed to acquire, process, analyze, and display indoor lighting and air-conditioning performance data, as well as generate automated assessments and recommendations.

***Commercial Lux Meter*.** A commercial lux meter, such as the Extech LT300 or its equivalent, shall be used as the reference instrument for validating the illumination measurements obtained by the developed system during the accuracy evaluation.

Digital Multimeter and Clamp Meter. A digital multimeter and clamp meter, such as the Fluke 117 and Fluke 323 or their electrical equivalents, shall be used as the reference instruments for validating the voltage and current measurements acquired by the developed system.

***Digital Thermometer*.** A digital thermometer, such as the Testo 925, HTC-1, or its equivalent, shall be used as the reference instrument for validating the room air temperature and supply-air temperature measurements obtained by the developed system.

***Commercial Anemometer*.** A commercial digital anemometer, such as the Extech AN100, Testo 410i, or its equivalent, shall be used as the reference instrument for validating the air velocity measurements acquired by the developed system. The measured air velocity shall also be used in determining the airflow rate and cooling capacity of the measured air-conditioning system.

Data to be Gathered

Illumination (lux). Illumination data, expressed in lux (lx), will be collected using the BH1750 digital light sensor installed in the developed system. The measured illumination levels will be compared with the readings obtained from a commercial lux meter to validate the accuracy of the lighting measuring module. The collected data will be used to determine whether the measured illumination complies with the recommended illumination levels for indoor spaces.

Voltage (V). Voltage measurements, expressed in volts (V), will be collected from the lighting and air-conditioning electrical circuits using the voltage sensor integrated into the developed system. The recorded values will be compared with the measurements obtained from a digital multimeter to determine the voltage measurement accuracy.

Current (A). Current measurements shall be gathered from the measured air-conditioning system to determine its electrical input and serve as one of the parameters for computing the electrical input power.

Room Temperature (°C). Room air temperature measurements shall be gathered to evaluate the indoor thermal condition and serve as one of the parameters for computing the cooling capacity of the measured air-conditioning system.

Supply-Air Temperature (°C). Supply-air temperature measurements shall be gathered at the air-conditioning outlet to determine the temperature difference required for computing the cooling capacity and evaluating the cooling performance of the measured air-conditioning system.

Air Velocity (m/s). Air velocity measurements shall be gathered at the air-conditioning outlet to determine the airflow rate and serve as one of the parameters for computing the cooling capacity and energy efficiency of the measured air-conditioning system.

Parameters to be Analysed

This section will describe the parameters that will be analyzed to evaluate the performance and energy efficiency of the proposed Smart Facility Measuring System for Indoor Lighting and Air-Conditioning. The collected parameters will be obtained through the developed measuring system and validated using reference measuring instruments. The analyzed parameters will serve as the basis for assessing lighting conditions, electrical consumption, air-conditioning performance, and overall system efficiency.

Percentage Error

The accuracy of the developed prototype shall be determined by comparing the measurements obtained from the prototype with those obtained from standard reference instruments using the percentage error equation.

$$\%\text{Error} = \left| \frac{M_p - M_r}{M_r} \right| \times 100$$

Where:

$\%\text{Error}$ = Percentage error (%)

M_p = Prototype measurement

M_r = Reference instrument measurement

Table 12.

Percentage Error Interpretation Scale

Percentage Error (%)	Interpretation
0-5	Highly Accurate
>5-10	Accurate
>10-15	Moderately Accurate
>15	Needs Improvement

Reliability

The reliability of the developed prototype shall be determined by evaluating its consistent operation across different testing locations using the reliability equation.

$$\text{Reliability } (\%) = \frac{N_p}{N_t} \times 100$$

Where:

Reliability (%) = Overall reliability of the developed prototype

N_p = Number of testing locations that successfully passed the reliability evaluation

N_t = Total number of testing locations evaluated

Table 13.

Reliability Rating Scale

Reliability (%)	Interpretation
90-100	Highly Reliable
80-89	Reliable
70-79	Moderately Reliable
Below 79	Needs Improvement

Table 14.

HVAC Energy Efficiency Assessment Criteria

Estimated Efficiency Retention	Assessment
≥95% of Rated Performance	Efficient
85-94%	Acceptable
70-84%	Needs Maintenance
<70%	Inefficient

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Appendix A

(Work Schedule)

ACITIVITIES	DURATION	MEMBER(S)	2026-2027																																					
			AUGUST				SEPTEMBER				OCTOBER				NOVEMBER				DECEMBER				JANUARY				FEBRUARY				MARCH				APRIL						
			1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
DESIGN PHASE																																								
Preparation of system design specifications	08/03/26 - 08/14/26	FC, ES, M, JK																																						
Development of the indoor lighting monitoring module design	08/10/26 - 08/28/26	FC, ES, M, JK																																						
Development of the air-conditioning monitoring module design	08/17/26 - 09/04/26	FC, ES, M, JK																																						
Integration of the mobile and desktop application framework	08/31/26 - 09/18/26	FC, ES, M, JK																																						
Formulation of testing and evaluation procedures	09/14/26 - 09/30/26	FC, ES, M, JK																																						
CONSTRUCTION PHASE																																								
Canvassing of electronic components and materials	10/01/26 - 10/09/26	FC, ES, M, JK																																						
Buying of selected components	10/05/26 - 10/16/26	FC, ES, M, JK																																						
Coordination with the fabricator	10/12/26 - 10/23/26	FC, ES, M, JK																																						
Assembly of the prototype hardware	10/19/26-11/20/26	FC, ES, M, JK																																						
Programming of the monitoring system	10/26/26 - 11/27/26	FC, ES, M, JK																																						
Integration of hardware and software modules	11/16/26 - 12/11/26	FC, ES, M, JK																																						
Debugging of the developed prototype	12/07/26 - 12/18/26	FC, ES, M, JK																																						
TESTING PHASE																																								
Functionality Testing	01/04/27 - 01/15/27	FC, ES, M, JK																																						
Validation of sensor accuracy	01/11/27 - 01/29/27	FC, ES, M, JK																																						
Calibration of measuring devices	01/18/27 - 01/29/27	FC, ES, M, JK																																						
Evaluation of system reliability	02/01/27 - 02/12/27	FC, ES, M, JK																																						
Collection of experimental data	02/08/27 - 02/19/27	FC, ES, M, JK																																						
Analysis of research findings	02/15/27 - 02/26/27	FC, ES, M, JK																																						
Finalization of research manuscript	02/15/27 - 02/26/27	FC, ES, M, JK																																						
FINAL DEFENSE																																								
Preparation for final defense	02/22/27 - 03/05/27	FC, ES, M, JK																																						
Final defense	03/05/2027	FC, ES, M, JK																																						
HARD BOUND	04/02/27 - 04/05/27	FC, ES, M, JK																																						

Appendix B

Budget Plan

Description	Quantity	Unit Cost	Cost
ESP32 Development Board	2	500.00	1000.00
BH1750 Light Intensity Sensor	4	180.00	720.00
AC Voltage Sensor Module	1	300.00	300.00
AC Current Sensor (ACS712/Equivalent)	1	350.00	350.00
DS18B20 Temperature Sensor	2	200.00	400.00
Digital Anemometer / Air Velocity Sensor	1	2500.00	2500.00
Breadboards / PCB	2	250.00	500.00
Jumper Wires and Connectors	1 set	500.00	500.00
Power Adapter / Power Supply	2	350.00	700.00
Electrical Accessories	1 set	500.00	500.00
Acrylic / Prototype Enclosure	1	1500.00	1500.00
Hardware Subtotal			**8970.00**
Prototype Making			2000.00
Assembly Materials			800.00
Mechanical Mounting Materials			700.00
Prototype Finishing			500.00
Prototype Subtotal			**4000.00**
Rental/Borrowing of Reference Instruments			1,500.00
Calibration Materials			500.00
Internet/Data for Wireless Testing			500.00
Testing Subtotal			**2,500.00**
Printing			800.00
Hardbound Copies			1,000.00
Subtotal			**1,800.00**
		Total Cost	**17,270.00**